



MATHEMATICS FOR COMPUTING

Assignment - 1

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Sets & Operations

Set Theory

Definition

A **set** is a well-defined collection of distinct objects. The objects that belong to a set are called its *elements* or *members*.

Set-Builder Notation

Instead of listing every element, a set can be described by stating the properties that its members must satisfy:

$$S = \{x \mid P(x)\} \quad \text{or} \quad S = \{x : P(x)\}$$

Read as: "The set of all x such that property $P(x)$ holds true."

Set Theory

Roster vs. Set-Builder Form

- **Roster (Tabular) Form:** Elements are explicitly listed inside curly braces.

$$A = \{2, 4, 6, 8, 10\}$$

- **Set-Builder Form:** Described by a condition.

$$A = \{x \in \mathbb{Z}^+ \mid x \text{ is even and } 2 \leq x \leq 10\}$$

Common Mathematical Sets

- $\mathbb{N} = \{1, 2, 3, \dots\}$ *(Set of Natural Numbers)*
- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ *(Set of Integers)*
- $\mathbb{Q} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\}$ *(Set of Rational Numbers)*

Types of Sets

Types of Sets

1. Null (Empty) Set ($\emptyset$)

Definition

A set containing no elements, with cardinality $|\emptyset| = 0$.

Roster Form

$$A = \{\}$$

Set-Builder Form

$$A = \{x \in \mathbb{R} \mid x^2 + 1 = 0\}$$

AI / Data Science Example

Set of invalid predictions with confidence score greater than 1:

$$S = \{p \in \text{Predictions} \mid \text{Confidence}(p) > 1.0\} = \emptyset$$

Types of Sets

2. Singleton Set

Definition

A set containing exactly one element, with cardinality $|A| = 1$.

Roster Form

$$B = \{0\}$$

Set-Builder Form

$$B = \{x \in \mathbb{Z} \mid x + 5 = 5\}$$

AI / Data Science Example

Target class assigned to a single image in multi-class classification:

$$Y = \{y \in \text{Classes} \mid y = \text{"Heart Murmur"}\}$$

Types of Sets

3. Finite Set

Definition

A set with a countable, limited number of elements ($|A| < \infty$).

Roster Form

$$C = \{2, 4, 6, 8, 10\}$$

Set-Builder Form

$$C = \{x \in \mathbb{Z}^+ \mid x \text{ is even} \wedge x \leq 10\}$$

AI / Data Science Example

Set of hyperparameter learning rates evaluated during grid search:

$$L = \{\eta \in \mathbb{R} \mid \eta = 10^{-k}, k \in \{1, 2, 3, 4\}\}$$

Types of Sets

4. Infinite Set

Definition

A set with unlimited elements whose counting process does not end ($|A| = \infty$).

Roster Form

$$D = \{1, 2, 3, 4, \dots\}$$

Set-Builder Form

$$D = \{x \in \mathbb{R} \mid 0 \leq x \leq 1\}$$

AI / Data Science Example

Continuous weight matrix parameter space of a dense layer:

$$W = \{w \in \mathbb{R}^{d_{in} \times d_{out}} \mid -1.0 \leq w \leq 1.0\}$$

Types of Sets

5. Countable Set

Definition

A set that can be placed in a one-to-one correspondence with natural numbers $\mathbb{N}$.

Roster Form

$$E = \{1, 3, 5, 7, \dots\}$$

Set-Builder Form

$$E = \{x \in \mathbb{N} \mid x = 2k - 1, k \in \mathbb{N}\}$$

AI / Data Science Example

Sequence of training epoch iterations in model optimization:

$$T = \{e \in \mathbb{N} \mid 1 \leq e \leq \text{MaxEpochs}\}$$

Types of Sets

6. Subset ($A \subseteq B$)

Definition

Set A is a subset of B if every element in A is also contained in B .

Roster Form

$$A = \{1, 2\}, \quad B = \{1, 2, 3, 4\}$$

Set-Builder Form

$$A = \{x \in B \mid P(x) \text{ is true}\}$$

AI / Data Science Example

Validation split taken directly from full training dataset:

$$D_{\text{val}} = \{d \in D_{\text{total}} \mid \text{IsValid}(d) = \text{true}\}$$

Types of Sets

7. Superset ($B \supseteq A$)

Definition

Set B is a superset of A if B contains all elements of A along with additional elements.

Roster Form

$$B = \{1, 2, 3, 4, 5\}, \quad A = \{2, 4\}$$

Set-Builder Form

$$B = \{x \mid x \in A \vee x \in A^c\}$$

AI / Data Science Example

Complete dataset holding all cross-validation partitions combined:

$$D_{\text{total}} = \{x \mid x \in D_{\text{train}} \cup D_{\text{val}} \cup D_{\text{test}}\}$$

Types of Sets

8. Equal Sets ($A = B$)

Definition

Two sets A and B are equal if $A \subseteq B$ and $B \subseteq A$ (same elements).

Roster Form

$$A = \{1, 2, 3\}, \quad B = \{3, 2, 1\}$$

Set-Builder Form

$$A = \{x \in \mathbb{Z} \mid 1 \leq x \leq 3\}, \quad B = \{y \in \mathbb{Z} \mid 1 \leq y \leq 3\}$$

AI / Data Science Example

Identical feature sets derived from two duplicate raw data columns:

$$F_1 = \{x \in \text{Data} \mid \text{ID}(x)\} = F_2$$

Types of Sets

9. Equivalent Sets ($A \sim B$)

Definition

Two sets A and B are equivalent if they have the same cardinality ($|A| = |B|$).

Roster Form

$$A = \{a, b, c\}, \quad B = \{1, 2, 3\}$$

Set-Builder Form

$$A = \{x \in \mathbb{N} \mid x \leq 3\}, \quad B = \{y \in \mathbb{Z}^- \mid y \leq -1\}$$

AI / Data Science Example

Balanced binary classification dataset with equal positive and negative samples:

$$|D_{\text{pos}}| = |D_{\text{neg}}| = 500$$

Types of Sets

10. Power Set ($\mathcal{P}(S)$)

Definition

The set of all subsets of S . Its size is $|\mathcal{P}(S)| = 2^{|S|}$.

Roster Form ($S = \{1, 2\}$)

$$\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$$

Set-Builder Form

$$\mathcal{P}(S) = \{A \mid A \subseteq S\}$$

AI / Data Science Example

All possible feature combinations evaluated during exhaustive Feature Selection:

$$\mathcal{P}(F) = \{A \mid A \subseteq \{\text{Age, BMI, ECG}\}\}$$

Types of Sets

11. Partially Ordered Set (Poset)

Definition

A set S paired with a partial order relation $\leq$ that satisfies reflexivity, antisymmetry, and transitivity: $(S, \leq)$.

Roster / Subset Relation

Set-Builder Form

$$S = \{1, 2, 4\}, \quad a \leq b \iff a \text{ divides } b \quad P = \{(x, y) \in S \times S \mid x \leq y\}$$

$$R = \{(1, 1), (2, 2), (4, 4), (1, 2), (1, 4), (2, 4)\}$$

AI / Data Science Example

Hierarchy of sub-sampling layers or directed acyclic dependencies in neural networks:

$$(\mathcal{L}, \preceq) \quad \text{where } L_i \preceq L_j \text{ if Layer } L_i \text{ feeds into } L_j$$

Types of Sets

12. Hasse Diagram

Definition

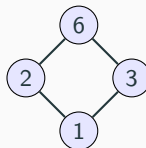
A visual representation of a finite partially ordered set (poset) that eliminates redundant reflexive self-loops and transitive edges.

Core Properties

- Directed upward (larger elements higher).
- Omits self-loops ($x \leq x$).
- Omits transitive edges ($a \leq b \wedge b \leq c \implies a \leq c$).

Example Poset D_6

$D_6 = \{1, 2, 3, 6\}$ under divisibility
($a \mid b$)



Inclusion-Exclusion Principle

Inclusion-Exclusion Principle

Principle of Inclusion-Exclusion (PIE)

Definition

A counting technique to compute the number of elements in the union of multiple sets without overcounting elements in their intersections.

Two-Set Formula

For any two finite sets A and B :

$$|A \cup B| = |A| + |B| - |A \cap B|$$

Three-Set Formula

For any three finite sets A , B , and C :

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |B \cap C| - |A \cap C| + |A \cap B \cap C|$$

Inclusion-Exclusion Principle

Example: Data Preprocessing & Cleaning

Problem Scenario

A audio dataset of 1000 recordings is checked for noise artifacts:

- Set A (High Noise): $|A| = 300$
- Set B (Clipping Artifacts): $|B| = 200$
- Set $A \cap B$ (Both High Noise and Clipping): $|A \cap B| = 80$

Calculation

Total unique corrupted audio files ($|A \cup B|$):

$$|A \cup B| = |A| + |B| - |A \cap B| = 300 + 200 - 80 = 420$$

Total clean audio files remaining = $1000 - 420 = 580$.

Examples of Set-Builder Form

1. Natural numbers less than 8:

$$A = \{x \in \mathbb{N} \mid x < 8\} = \{1, 2, 3, 4, 5, 6, 7\}$$

2. Even numbers from 2 to 20:

$$B = \{2k \mid k \in \mathbb{N}, 1 \leq k \leq 10\} = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$$

3. Odd numbers less than 15:

$$C = \{x \in \mathbb{N} \mid x \text{ is odd}, x < 15\} = \{1, 3, 5, 7, 9, 11, 13\}$$

4. Prime numbers less than 7:

$$D = \{x \in \mathbb{N} \mid x \text{ is prime}, x < 7\} = \{2, 3, 5\}$$

5. Multiples of 4 less than 25:

$$E = \{4k \mid k \in \mathbb{N}, 4k < 25\} = \{4, 8, 12, 16, 20, 24\}$$

6. Squares of odd numbers up to 7:

$$F = \{x^2 \mid x \in \mathbb{N}, x \text{ is odd}, x \leq 7\} = \{1, 9, 25, 49\}$$

7. A set formed by adding 2 to natural numbers:

$$G = \{x + 2 \mid x \in \mathbb{N}\} = \{3, 4, 5, 6, \dots\}$$

8. Factorials up to 5!:

$$H = \{n! \mid n \in \mathbb{N}, 1 \leq n \leq 5\} = \{1, 2, 6, 24, 120\}$$

9. Set formed by $n^2 - 1$ for natural numbers:

$$I = \{n^2 - 1 \mid n \in \mathbb{N}\} = \{0, 3, 8, 15, \dots\}$$

10. Factors of 12:

$$J = \{x \in \mathbb{N} \mid 12 \pmod{x} = 0\} = \{1, 2, 3, 4, 6, 12\}$$